

Towards Low-Cost Risk Stratification for Medullary Thyroid Carcinoma: A Reproducible Machine-Learning Benchmark in Published MEN2/RET-Carrier Cohorts

Harnoor Kaur^a, Arjun Vijay Prakash^a, Shashwat Mishra^{a*}

^aCity Montessori School, Quality Building, Sector G, LDA Colony, Kanpur Road, Lucknow, Uttar Pradesh 226012, India.

E-mail addresses: har.noor16@gmail.com (H. Kaur); arjunv.prakash12345@gmail.com (A. V. Prakash); mishra.shashwat4002@gmail.com (S. Mishra).

Corresponding author: Shashwat Mishra, City Montessori School, Quality Building, Sector G, LDA Colony, Kanpur Road, Lucknow, Uttar Pradesh 226012, India. E-mail: mishra.shashwat4002@gmail.com. Telephone: to be supplied before journal upload. Fax: not available.

Abstract

Multiple Endocrine Neoplasia type 2 (MEN2) is a rare hereditary endocrine cancer syndrome. RET genetic testing is central to care, but it may be cost-prohibitive in resource-constrained settings. This study developed a reproducible machine-learning benchmark for lower-cost medullary thyroid carcinoma (MTC) risk stratification among published MEN2/RET-carrier records. We curated 149 RET carriers from 10 peer-reviewed sources spanning 14 RET variants. Demographic, genotype, clinical, and biomarker variables were harmonized, and logistic regression, random forest, XGBoost, LightGBM, and linear support vector machine models were evaluated under an 80/20 stratified internal held-out split. The main triage-oriented analysis was a genotype-blind XGBoost model that removed RET variant and American Thyroid Association (ATA) risk features; it retained 93.3% sensitivity with 73.3% accuracy. As a sequencing-informed comparator, the genotype-aware XGBoost model reached high sensitivity (100.0%) with 83.3% accuracy. A secondary synthetic-augmentation experiment reached 96.2% accuracy with LightGBM, but it is interpreted only as a simulation because synthetic controls may not represent real clinical heterogeneity. This is not a deployable diagnostic tool. It is a transparent rare-disease benchmarking workflow providing preliminary evidence that routine clinical and biomarker features may support high-sensitivity triage in future prospective studies, while exposing feature-dependence, imputation uncertainty, and biomarker-timing limitations. Independent prospective validation is required before clinical or screening use.

Keywords

MEN2; RET; Medullary thyroid carcinoma; Calcitonin; Carcinoembryonic antigen; Rare disease; Machine learning; Screening; Risk stratification; Reproducibility

1 Introduction

Multiple Endocrine Neoplasia type 2 (MEN2) is a hereditary endocrine cancer syndrome caused by germline pathogenic variants in the RET proto-oncogene. MEN2 is clinically associated with medullary thyroid carcinoma (MTC), and may also involve pheochromocytoma and primary hyperparathyroidism depending on the MEN2 subtype and RET variant. Current clinical management is guided by genotype-phenotype correlations, serum calcitonin and carcinoembryonic antigen (CEA), family screening, and guideline-based timing of thyroidectomy or surveillance (1).

Computational modeling in MEN2 is limited by the rarity of the syndrome. Related machine-learning work in MTC has focused mainly on post-diagnosis tasks such as metastatic risk, recurrence, or prognosis, including a model for lateral lymph-node metastasis that used clinical baseline characteristics and TI-RADS features (22). To our knowledge, no prior study has developed a reproducible MEN2/RET-carrier benchmark specifically testing whether routine clinical and biomarker features can support high-sensitivity triage before genetic sequencing.

The practical motivation for this research is rooted in healthcare equity. In India, MEN2-related RET genetic testing is advertised at about INR 20,000, or approximately 225 USD, which can put life-saving diagnosis out of reach for many families (19). By contrast, online Indian

laboratory listings place serum calcitonin and CEA assays at roughly INR 700 each, although prices vary by city, laboratory, and collection model (20,21). A highly sensitive biomarker-driven triage model could therefore help identify which patients or families most urgently need referral for genetic sequencing. This research asks: *can machine learning on routine blood biomarkers, including calcitonin and CEA, and clinical features stratify MTC risk without using expensive genetic sequencing as an input?* The available literature is scientifically valuable for asking this question. It also creates methodological risks for machine learning: small sample size, overlapping familial patterns, inconsistent biomarker reporting, post-diagnostic measurements, missing CEA values, and possible circularity when genotype-derived risk categories are used to predict cancer status.

This study asks a narrower question than clinical diagnosis: can published MEN2/RET-carrier data be converted into a reproducible benchmark that quantifies how model performance changes when genotype, biomarker, and synthetic-augmentation assumptions are varied? Ultimately, this project asks whether routine blood and clinical features contain useful structure for high-sensitivity triage in a rare cancer dataset. They cannot yet be used as a low-cost substitute for RET sequencing, clinical evaluation, or guideline-based care.

To address the main validity concern, the analysis separates two modeling tracks. The genotype-blind analysis removes RET variant and American Thyroid Association (ATA) risk features and is treated as the clinically motivated triage benchmark because it relies on non-genetic clinical and biomarker variables. The genotype-aware benchmark includes RET variant and ATA-aligned risk features and is interpreted as a sequencing-informed comparator relevant only after genetic context is already known. A secondary synthetic-augmentation experiment is reported separately as a simulation, not as an expanded real-world cohort.

2 Materials and Methods

2.1 Study Design

This was a retrospective secondary analysis of published, de-identified MEN2/RET-carrier records. No new patients were recruited, and no new biological specimens were collected. The primary dataset was the literature-derived cohort of 149 confirmed RET carriers. The target variable was binary MTC status: MTC present vs. MTC absent in the extracted source record.

The sample size should be interpreted in the context of MEN2 rarity. The 149-carrier cohort is not large by common machine-learning standards, but it represents a deliberately curated rare-disease benchmark assembled from 10 peer-reviewed sources, 14 RET variants, and both MTC-positive and MTC-negative carrier records. The goal was therefore not to claim population-level generalizability from a small dataset. It was to create an auditable carrier-level dataset large enough to test sensitivity-oriented benchmarks, expose failure modes, and guide prospective data collection.

Table 1: Baseline characteristics of the literature-derived cohort stratified by MTC status.

Characteristic	Overall (n=149)	MTC present (n=73)	MTC absent (n=76)
Age, mean years	34.27	43.30	25.59
Age, median years	31.5	43.0	20.0
Age range, years	0.17–90.0	0.17–85.0	1.0–90.0
Female sex, n (%)	99 (66.4)	46 (63.0)	53 (69.7)
Male sex, n (%)	50 (33.6)	27 (37.0)	23 (30.3)
Unique RET variants, n	14	11	13
Calcitonin elevated, n (%)	58 (38.9)	47 (64.4)	11 (14.5)
CEA imputed, n (%)	137 (91.9)	65 (89.0)	72 (94.7)
Thyroid nodules present, n (%)	10 (6.7)	8 (11.0)	2 (2.6)
Family history of MTC, n (%)	105 (70.5)	53 (72.6)	52 (68.4)
Pheochromocytoma, n (%)	9 (6.0)	9 (12.3)	0 (0.0)
Hyperparathyroidism, n (%)	5 (3.4)	4 (5.5)	1 (1.3)

2.2 Data Sources and Eligibility

Studies were eligible if they reported patient-level or reconstructable carrier-level information on RET-positive individuals, MTC status, age or demographic variables, and at least one clinical, genotype, or biomarker feature usable for harmonization. Sources were excluded if they lacked RET-carrier-level information, duplicated already extracted records, reported only aggregate non-reconstructable summaries, or focused on non-RET thyroid cancer without MEN2-carrier information.

Table 2: Published sources used to construct the literature-derived cohort.

Study	Source	Main variant context	n	DOI
1	Xu et al., 2016 (3)	RET K666N families	24	10.1089/thy.2016.0374
2	Frank-Raue et al., 2006 (2)	Multiple RET variants after thyroidectomy	46	10.1530/eje.1.02216
3	Qi et al., 2015 (5)	RET S891A/R525W familial MTC	15	10.18632/oncotarget.4992
4	Florescu et al., 2024 (8)	RET C634G kindred	6	10.3390/clinpract14060179
5	La Greca et al., 2024 (9)	RET K666N family	4	10.1530/EDM-24-0009
6	Vijayan et al., 2021 (7)	Indian RET S891A pedigree	7	10.4103/ijc.IJC_639_19
7	Zhang et al., 2024 (10)	RET C634Y family	3	10.12998/wjcc.v12.i15.2627

Study	Source	Main variant context	n	DOI
8	Shankar et al., 2012 (11)	Familial MEN2B infant	2	10.1186/1687-9856-2012-25
9	Schulte et al., 2010 (4)	RET S891A multi-center cohort	36	10.1210/jc.2010-0375
10	Qi et al., 2014 (6)	Chinese Han RET S891A family	6	10.1007/s12038-014-9428-x

2.3 Data Extraction and Harmonization

Each source was converted into structured JSON and then into a tabular dataset. Extracted variables included age, sex, RET variant, ATA-aligned risk level, calcitonin elevation and numeric value, CEA numeric value, CEA missingness flag, thyroid nodule indicators, family history of MTC, pheochromocytoma, hyperparathyroidism, MEN2 syndrome status, C-cell disease status, and MTC diagnosis. RET variants were normalized to protein-level short notation where possible.

The genotype-blind feature set included age, sex, calcitonin features, CEA features, thyroid nodule variables, family history of MTC, pheochromocytoma, hyperparathyroidism, MEN2 syndrome indicators, age group, and simple non-genetic interaction terms. The genotype-aware comparator additionally included RET variant one-hot encodings, ATA-aligned risk level, and ATA-risk interaction terms. The no-CEA sensitivity analysis removed CEA numeric values. The biomarker-timing sensitivity analysis removed calcitonin, CEA, and nodule-derived features, leaving genotype and demographic features.

Table 3: Key variable groups used in the benchmark.

Variable group	Variables
Demographic	Age, sex, age group, age squared
Genotype-aware	RET variant one-hot encodings, ATA-aligned risk level, risk-age and risk-calcitonin interactions
Biomarker	Calcitonin elevated flag, numeric calcitonin, CEA numeric value, CEA missingness/imputation flag
Clinical presentation	Thyroid nodule indicators, family history of MTC, pheochromocytoma, hyperparathyroidism
Outcome	Binary MTC status extracted from source reports

2.4 Missing Data and CEA Imputation

CEA was observed in only 12 of 149 records before imputation. MICE with predictive mean matching was used as one exploratory imputation strategy and compared with no-CEA and simpler imputation alternatives in sensitivity analyses (12,23). CEA imputation was not treated as independently validated because the observed paired calcitonin-CEA subset was small and weakly correlated ($r = 0.1158$, $n = 12$). A binary CEA-imputed flag was retained to preserve missingness information.

Table 4: Selected missingness and harmonization summary in the 149-record literature-derived cohort.

Field	Observed or nonzero	Total	Note
Age	149	149	Missing ages were filled from source-level medians where required.
Sex	149	149	Unknown values were mode-filled for modeling.
RET variant	149	149	Required for genotype-aware inclusion.
Calcitonin numeric value	77	149	Zero includes unavailable, normal, or undetectable values after harmonization.
CEA observed before imputation	12	149	Observed paired CEA values seeded the imputation analysis.
CEA imputed	137	149	Missingness flag retained in the model dataset.

2.5 Biomarker Timing and Leakage Handling

Biomarker timing varied across sources. Some studies reported preoperative calcitonin, some reported surveillance values from cascade testing, and some case reports mixed diagnostic, preoperative, postoperative, or follow-up measurements. Because patient-level timing was not uniformly recoverable across all extracted records, the primary manuscript reports biomarker timing as a limitation and includes a conservative biomarker-exclusion sensitivity analysis.

2.6 Models and Statistical Approach

Five model families were benchmarked: logistic regression, random forest, XGBoost, LightGBM, and linear support vector machine. Implementations used scikit-learn, XGBoost, and LightGBM (14-16). The dataset was split into 80% training and 20% internal held-out testing using stratification and fixed random seed 42. Feature scaling was fit on training data only and then applied to the held-out split. SMOTE was applied only to training folds where class balancing was used (13). These controls reduce ordinary preprocessing leakage: the held-out records were not used to fit the scaler, and synthetic oversampling was never applied to the held-out test set. CEA imputation was label-free, retained a missingness flag, and was evaluated with no-CEA sensitivity analyses. Performance metrics included accuracy, precision, sensitivity, specificity, F1-score, ROC-AUC, and average precision. Raw confusion-matrix counts were reported with Wilson 95% confidence intervals for binomial proportions so that small held-out sample sizes were visible. The benchmark was reported with emphasis on transparent prediction-model methodology rather than immediate clinical deployment (24).

The primary endpoint was internal held-out sensitivity of the genotype-blind model on the literature-derived cohort, reflecting the intended role of a future screening or triage tool in

Table 5: Source-level biomarker timing classification used for leakage assessment.

Study	Timing class	Leakage relevance
1	Mixed	Probands often had diagnostic/preoperative or follow-up values; relatives had cascade-surveillance values.
2	Preoperative	Basal/stimulated calcitonin before thyroidectomy; CEA not reported.
3	Preop./surv.	Family screening and surgery workup values; CEA mostly not individually reported.
4	Surv./preop.	Kindred-level MEN2A evaluation; CEA not central.
5	Preop./surv.	Family case-report values around diagnosis and carrier follow-up.
6	Surv./preop.	Pedigree screening values around family evaluation.
7	Post-diagnostic/Mixed	Small family report with values around established MEN2A diagnosis and surgery.
8	Mixed	Infant preoperative calcitonin plus postoperative/surveillance follow-up; mother post-diagnostic values.
9	Preoperative	Multicenter S891A cohort reports preoperative calcitonin multiples; CEA measured in subset.
10	Preop./postop.	Pedigree table includes pre- and postoperative calcitonin; extraction favored preoperative values where present.

which false negatives are especially undesirable. Secondary endpoints were accuracy, specificity, ROC-AUC, genotype-aware comparator performance, no-CEA performance, biomarker-exclusion performance, and performance in the synthetic-augmentation simulation. SHAP was used only as an exploratory interpretation method for supported models (17).

2.7 Synthetic-Augmentation Experiment

A secondary case-control simulation expanded the data to 1,047 records using variant-matched synthetic controls and synthetic MTC-like examples generated from source distributions, followed by SMOTE only within training data. These records are not real patients. Results from this experiment are therefore interpreted as a stress test of model behavior under augmentation, not as evidence of external clinical validity.

3 Results

3.1 Cohort Summary

The literature-derived cohort contained 149 RET carriers from 10 studies, with 73 records with MTC and 76 without MTC. Baseline demographic and clinical characteristics stratified by MTC status are summarized in Table 1. Fourteen RET variants were represented: S891A, K666N, C634Y, C634R, R525W, C634G, C630R, C634W, L790F, V804M, Y791F, C618S, C620Y, and M918T.

Figure 1 shows the RET variant distribution, Figure 2 shows the age distribution, and Figure 3 shows the limited paired calcitonin-CEA data used for exploratory CEA imputation.

3.2 Primary Genotype-Blind Triage Benchmark

In the primary internal held-out split of the 149-record literature-derived cohort, the genotype-blind XGBoost model produced a confusion matrix of TN = 8, FP = 7, FN = 1, and TP = 14. This corresponds to 73.3% accuracy (22/30; 95% Wilson interval 55.6%-85.8%) and 93.3% sensitivity (14/15; 95% Wilson interval 70.2%-98.8%). The model did not use RET variant or ATA risk features, making it the analysis most aligned with a low-cost triage scenario before genetic sequencing. Its high sensitivity is encouraging for screening-oriented risk stratification, but the single false negative and wide confidence interval show that it is not ready for clinical deployment.

3.3 Sequencing-Informed Comparator

In the same held-out split, the genotype-aware XGBoost model produced a confusion matrix of TN = 10, FP = 5, FN = 0, and TP = 15. This corresponds to 83.3% accuracy (25/30; 95% Wilson interval 66.4%-92.7%), high sensitivity (100.0%; 95% Wilson interval 79.6%-100.0%), and 66.7% specificity (10/15; 95% Wilson interval 41.7%-84.8%). The point estimate had no false negatives in this one split, but because the model uses RET variant and ATA-aligned risk features, it should be interpreted as a sequencing-informed benchmark rather than a low-cost triage solution.

Table 6: Internal held-out performance with raw counts and Wilson confidence intervals.

Analysis	Model	Confusion matrix	Sensitivity	Accuracy
Genotype-aware	XGBoost	TN=10, FP=5, FN=0, TP=15	100.0% (15/15; 79.6%-100.0%)	83.3% (25/30; 66.4%-92.7%)
Genotype-blind	XGBoost	TN=8, FP=7, FN=1, TP=14	93.3% (14/15; 70.2%-98.8%)	73.3% (22/30; 55.6%-85.8%)
No CEA	XGBoost	TN=12, FP=3, FN=0, TP=15	100.0% (15/15; 79.6%-100.0%)	90.0% (27/30; 74.4%-96.5%)
No biomarkers	XGBoost	TN=12, FP=3, FN=0, TP=15	100.0% (15/15; 79.6%-100.0%)	90.0% (27/30; 74.4%-96.5%)
Synthetic	LightGBM	TN=156, FP=3, FN=5, TP=46	90.2% (46/51; 79.0%-95.7%)	96.2% (202/210; 92.7%-98.1%)

3.4 Feature Sensitivity Analyses

Comparing the genotype-blind and genotype-aware XGBoost models suggests that non-genetic clinical and biomarker features may contain useful triage information, while genetic context still improves apparent performance in this small split. Therefore, the genotype-aware result should not be presented as a substitute for genetic testing. Instead, the genotype-blind result is best interpreted as an early benchmark for future triage systems that would flag high-risk patients for confirmatory genetic evaluation.

Removing CEA improved the XGBoost result in the original split to 90.0% accuracy while preserving high sensitivity (100.0%). This supports treating CEA imputation as exploratory rather than essential to the main result. Removing calcitonin, CEA, and nodule-derived features as a conservative biomarker-timing sensitivity analysis also preserved high sensitivity (100.0%) in this split, but this should not be interpreted as evidence that biomarkers are clinically unnecessary. It only shows that this small internal test split was not driven entirely by biomarker timing.

3.5 Synthetic-Augmentation Simulation

In the synthetic-augmentation experiment, LightGBM achieved 96.2% accuracy (202/210) and 90.2% sensitivity (46/51) on the internal held-out split. These values were higher than the corresponding LightGBM result on the original 149-record cohort, but the augmented dataset contains synthetic records and synthetic controls. The result is therefore useful for studying how augmentation changes discrimination, not for estimating real-world clinical performance.

3.6 Interpretability

Exploratory SHAP analysis was generated for supported models. In the synthetic LightGBM analysis, genotype and biomarker-related features both contributed to predictions. Because the synthetic analysis may reflect assumptions used to generate controls, SHAP rankings from that experiment should not be treated as clinical feature rankings.

4 Discussion

This study presents a reproducible benchmark for machine-learning analysis of MTC status in published MEN2/RET-carrier records, with the central clinical question reframed around triage toward low-cost access. The novel contribution is not a single high-performing classifier. It is the combination of carrier-level literature data collection, genotype-aware versus genotype-blind benchmarking, ablation analyses, imputation sensitivity analyses, and a reproducible pipeline that makes each assumption inspectable. In that context, the genotype-blind XGBoost model provides preliminary evidence that non-genetic clinical and biomarker variables may support high-sensitivity triage in resource-constrained settings.

The 149-carrier sample should not be dismissed simply because it is small. MEN2 is rare, and most available studies are family reports, case series, or variant-specific cohorts. This dataset consolidates those scattered records into a balanced carrier-level benchmark with 73 MTC-positive and 76 MTC-negative records across 14 RET variants. That structure is sufficient for an internal sensitivity benchmark, especially because the manuscript reports raw counts and Wilson intervals. It is not sufficient for clinical deployment, which is why the conclusion is framed as preliminary evidence requiring prospective validation.

The most important methodological issue is the role of genotype. RET variant and ATA risk level are clinically meaningful variables, but they can also encode prior knowledge about disease risk. The genotype-aware model is therefore relevant only for confirmed or suspected RET-carrier datasets where genetic context is already known. Its high sensitivity (100.0%) is useful as a sequencing-informed comparator, but it cannot support the low-cost triage argument because a patient with a known RET variant has already undergone genetic testing. The genotype-blind model better matches the healthcare-equity motivation: it asks whether routine blood tests and clinical features can identify people who most need sequencing, referral, or specialist review.

High internal performance estimates in this study should be interpreted as benchmark results, not as inflated clinical claims. Several steps reduce direct data leakage: genotype-blind models remove RET variant and ATA-risk inputs, feature scaling is learned only from training data, SMOTE is applied only to training data, and the held-out test set contains real records rather

than oversampled records. The remaining concerns are different: small-sample optimism, familial clustering, incomplete biomarker timing, and synthetic-control assumptions. These are why the manuscript emphasizes ablation analyses, no-CEA and biomarker-exclusion sensitivity analyses, and future prospective validation rather than deployment.

A future validated version of this workflow could help democratize rare-disease screening in resource-constrained settings. For example, a rural clinician could enter age, sex, calcitonin, CEA, thyroid nodule status, and family-history information into a simple decision-support application. A high-risk output would not diagnose MEN2 or MTC; instead, it would support referral to an endocrine oncology center, prioritization for RET sequencing, or counseling about the need for specialist evaluation. A low-risk output would still require clinical judgment, especially in families with known MEN2, but could help reduce unnecessary travel and testing when specialist access is limited.

The biology of low-cost biomarker triage also requires clinical nuance. Serum calcitonin is highly sensitive for MTC-related C-cell disease, but false-positive elevations can occur in settings such as chronic kidney disease, autoimmune thyroiditis, and smoking. Future triage models should therefore collect these confounders prospectively rather than treating calcitonin as a stand-alone diagnostic signal.

CEA imputation is another major limitation. Only 12 observed paired CEA values were available, and the observed calcitonin-CEA correlation was weak. MICE with predictive mean matching is a reasonable exploratory imputation method because PMM constrains imputed values toward observed donor values, but the no-CEA sensitivity analysis is more important for credibility than the imputed CEA result (12,23). Future versions should either collect more observed CEA values or treat CEA as optional/missing rather than central.

Biomarker timing also limits interpretation. Some calcitonin and CEA values were preoperative or surveillance measurements, while others were post-diagnostic or mixed with follow-up information. If a biomarker was measured after cancer diagnosis or surgery, it could leak outcome information into the model. Because timing could not be standardized across all records, the manuscript reports a conservative biomarker-exclusion sensitivity analysis and treats all biomarker-based findings as provisional.

The synthetic-augmentation experiment should be interpreted cautiously. Synthetic controls can make model training easier and improve apparent discrimination, but they may also encode assumptions about what non-MTC carriers look like. For this reason, the augmented LightGBM result is presented as a simulation of augmentation behavior, not as an expanded clinical cohort.

Several biological observations remain relevant for future work. RET K666N can show heterogeneous penetrance, and allele dosage may matter in some reports (3,9). RET S891A has been reported in Indian and Chinese pedigrees and may show variable clinical behavior across families (5-7). Codon 634 variants are associated with MEN2A phenotypes and have been reported with cutaneous lichen amyloidosis in some kindreds (8,10). These examples show why simple binary variant encoding is likely insufficient for future clinical modeling.

4.1 Limitations

The main limitations are small sample size, retrospective literature-derived data, heterogeneous source reporting, possible familial clustering, incomplete biomarker timing, extensive CEA imputation, and lack of independent external validation. The held-out split is internal to the curated

literature dataset and does not test generalization to a new institution, new country, or prospective clinical population. The public dataset was reconstructed from published reports; although no new identifiers were collected, rare-disease pedigrees can carry re-identification risk. Public-facing datasets should therefore use anonymized row identifiers and avoid unnecessary family labels or exact relationship details.

5 Conclusion

MEN2 Predictor is best framed as a necessary first-step benchmark toward equitable rare-disease triage, not as a deployable diagnostic system. The genotype-blind model provides preliminary evidence that routine clinical and biomarker variables may support high-sensitivity MTC risk stratification without using genetic sequencing as an input, while the genotype-aware model establishes a sequencing-informed comparator. The workflow shows how performance changes when genotype features, CEA imputation, biomarker-dependent features, and synthetic augmentation are varied. These results support the establishment of prospective, multi-center MEN2 registries in India and other resource-constrained settings to gather standardized, pre-diagnostic biomarker and clinical-confounder data before clinical or screening use.

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Author Contributions

H.K. contributed to literature search, study identification, data curation, and biological interpretation. A.V.P. designed and implemented the machine-learning pipeline, trained the models, and performed the computational analysis. S.M. provided mentorship, advised on methodological decisions, and oversaw the project direction. All authors reviewed and approved the final manuscript.

Conflict of Interest

The authors declare no conflicts of interest.

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Data Availability

The analysis code, processed data artifacts, and figure-generation scripts are available in the GitHub repository at <https://github.com/arjuncodess/men2-predictor>. Anonymized submission copies of the processed original and augmented datasets are provided under `paper/submission_data/` to reduce avoidable patient- or family-level identifiability in review materials; repository materials should be reviewed before public submission to confirm that no unnecessary family labels, exact relationship details, or other potentially identifying rare-disease metadata are exposed. The hosted Hugging Face Space is provided for educational demonstration only and must not be used for diagnosis, treatment, triage, or any real-world clinical decision.

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Table 7: RET-carrier counts by source study label used in the aggregated cohort (total n=149).

Study label	Patients (n)
European Journal of Endocrinology (2006)	46
JCEM (2010) RET S891A MEN2A Spectrum	36
Thyroid Journal (2016)	24
Oncotarget (2015) RET S891A FMTC	15
Indian Journal of Cancer (2021) RET S891A Family	7
Clinics and Practice (2024) RET C634G Kindred	6
Journal of Biosciences (2014) RET S891A Chinese FMTC Family	6
Endocrinol. Diabetes Metab. Case Reports (2024) RET K666N	4
World Journal of Clinical Cases (2024) RET C634Y Family	3
Int. J. Pediatr. Endocrinol. (2012) Familial MEN2B Infant	2

Table 8: Ablation study results: XGBoost on the original paper-only dataset (n=149). Accuracy change is relative to baseline.

Configuration	Features	Accuracy	Recall	F1	Δ Acc
Baseline (All Features)	33	0.833	1.000	0.857	—
Without ATA Risk Level	30	0.767	0.867	0.788	−6.7%
Without Variant Encodings	19	0.833	1.000	0.857	0.0%
Without Any Genetic Features	16	0.733	0.933	0.778	−10.0%
Without Calcitonin	30	0.833	1.000	0.857	0.0%
Without CEA	32	0.900	1.000	0.909	+6.7%
Genetics Only (No Biomarkers)	27	0.900	1.000	0.909	+6.7%
Biomarkers Only (No Genetics)	16	0.733	0.933	0.778	−10.0%

Table 9: Ablation study results: LightGBM on the expanded dataset (n=1,047). Accuracy change is relative to baseline.

Configuration	Features	Accuracy	Recall	F1	Δ Acc
Baseline (All Features)	34	0.962	0.902	0.920	—
Without ATA Risk Level	31	0.952	0.882	0.900	−1.0%
Without Variant Encodings	20	0.943	0.863	0.880	−1.9%
Without Any Genetic Features	17	0.933	0.824	0.857	−2.9%
Without Calcitonin	31	0.952	0.882	0.900	−1.0%
Without CEA	33	0.929	0.882	0.857	−3.3%
Genetics Only (No Biomarkers)	27	0.910	0.843	0.819	−5.2%
Biomarkers Only (No Genetics)	17	0.933	0.824	0.857	−2.9%